

EdgeBench: A Lightweight Preprocessing and Streaming Ingestion Framework for Edge AI Pipelines



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Motivation and Objective

- **Ingest Problem:** Continuous instrument streams can overwhelm edge systems.
- **Compute Mismatch:** ML workloads are often too heavy for low-power edge nodes.
- **Objective:** Develop a lightweight preprocessing and ingestion framework for downstream classification.

Contributions

- Portable streaming ingestion framework
- Bounded-state preprocessing pipeline
- Cross-platform telemetry that exposes runtime bottlenecks

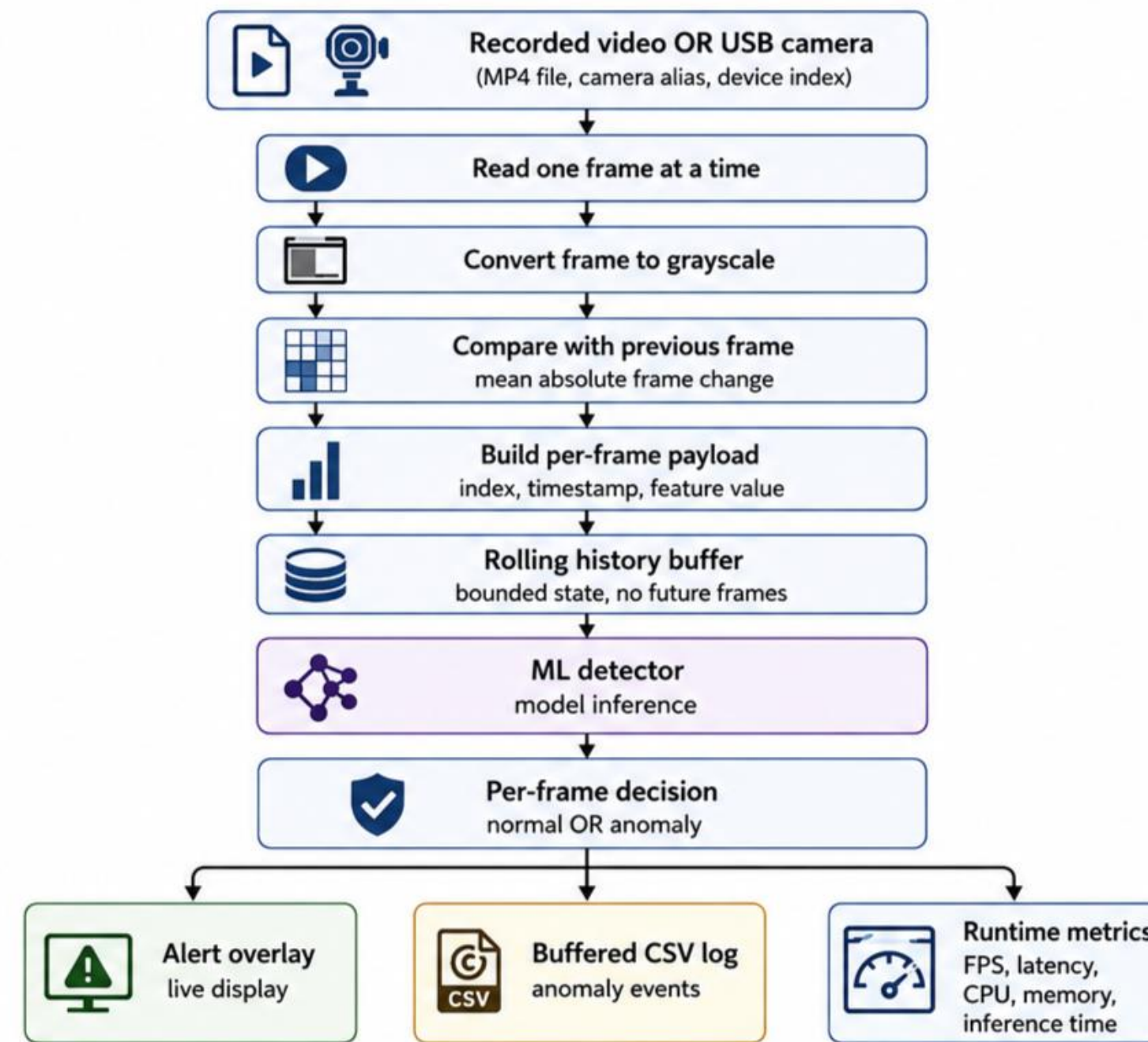
Experimental Evaluation

- **Heterogenous Platform Tested:** High-compute CPU is 12th Gen Intel(R) Core (TM) i7-12700H Workstation vs. Low-power Embedded ARM-based (ARM Cortex-A9) Xilinx PYNQ Platform
- **Workload Modalities:** Prerecorded MP4 file loops vs. Live Physical USB 2.0 Camera streams

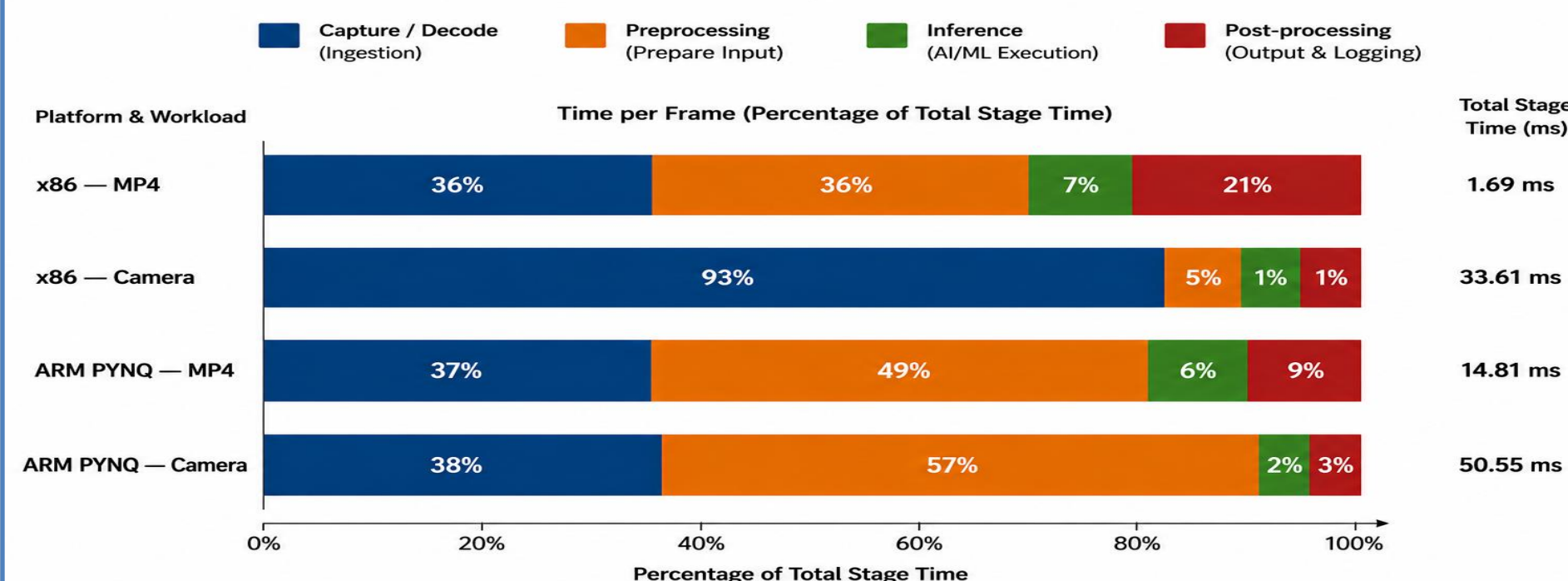
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EdgeBench Framework



Internal Pipeline Timing Breakdown (Execution Time Per Frame)



- Percentage indicated the contribution of each stage to the total stage time
- Lower total stage time indicated better performance

Overall Benchmark Performance

Platform & Workload	Latency / Frame	Throughput	Peak Memory RSS
x86 Workstation (MP4)	0.76 ms	574.24 FPS	219.78 MB
x86 Workstation (Camera)	1.76 ms	29.72 FPS	197.26 MB
ARM Xilinx PYNQ (MP4)	8.22 ms	66.35 FPS	147.43 MB
ARM Xilinx PYNQ (Camera)	29.98 ms	19.71 FPS	143.09 MB

Conclusions and Key Findings

- **Portable:** Common pipeline runs across x86 and ARM/PYNQ hardware.
- **Modular:** Unified ingestion, bounded-state processing, pluggable inference, and runtime telemetry.
- **Low-Overhead:** Buffered processing and telemetry support continuous streaming with stable memory usage.
- **Bottleneck Exposed:** On PYNQ, capture and preprocessing dominate inference time.
- **Key Insight:** Edge-AI performance can be constrained by front-end processing rather than inference alone.
- **Edge-to-HPC:** EdgeBench provides a preprocessing foundation for routing sensor streams toward HPC backends.
- **Future Work:** Extend validation to multimodal sensor streams.

References

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